Lecture 20

Waveform tomography

From last time…

Using Fermat's principle, we find that:

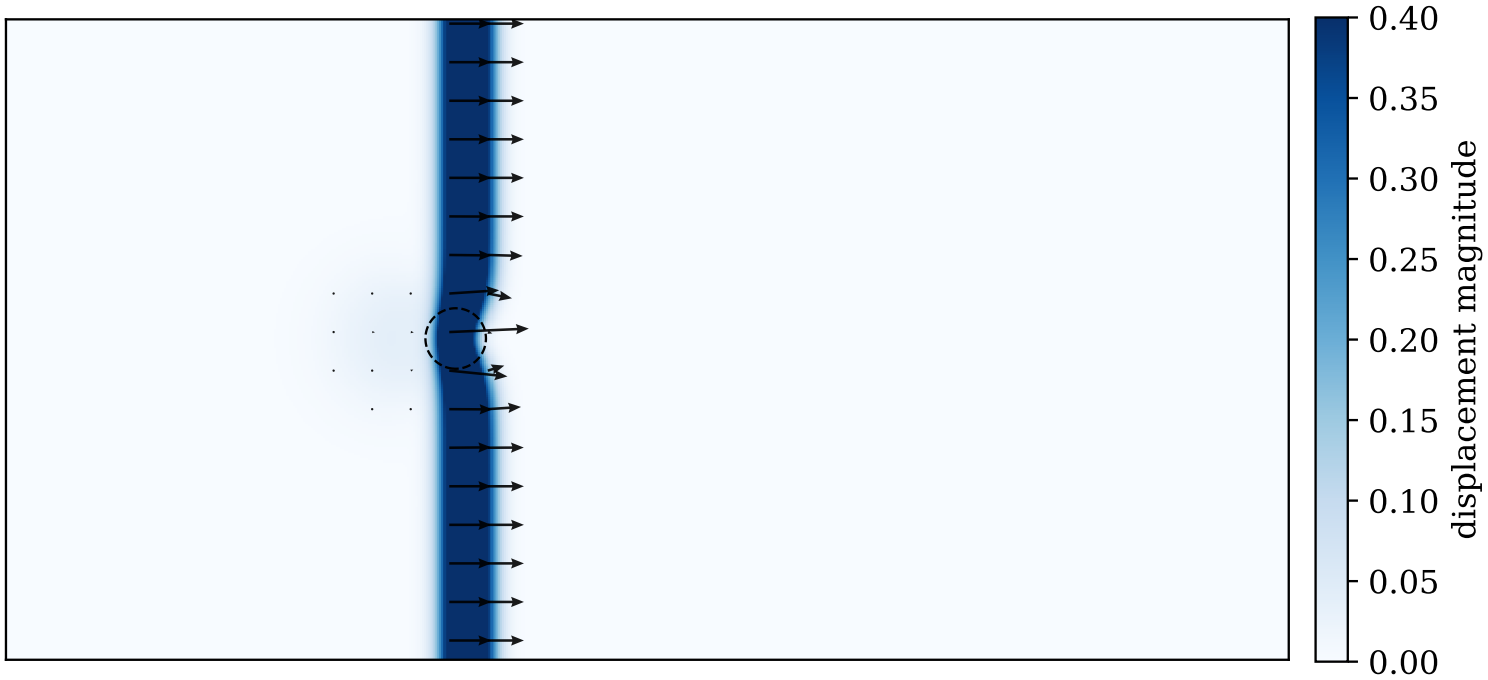
$$\delta T = -\int_{\text{ray}} \frac{\delta v}{v^2}\,\mathrm{d}s.$$

Expand the velocity in a finite-dimensional basis:

$$\delta\alpha(\mathbf{x}) = \sum_{j=1}^{m} \delta\alpha_j \varphi_j(\mathbf{x}).$$

Formulation as a regularised least squares problem:

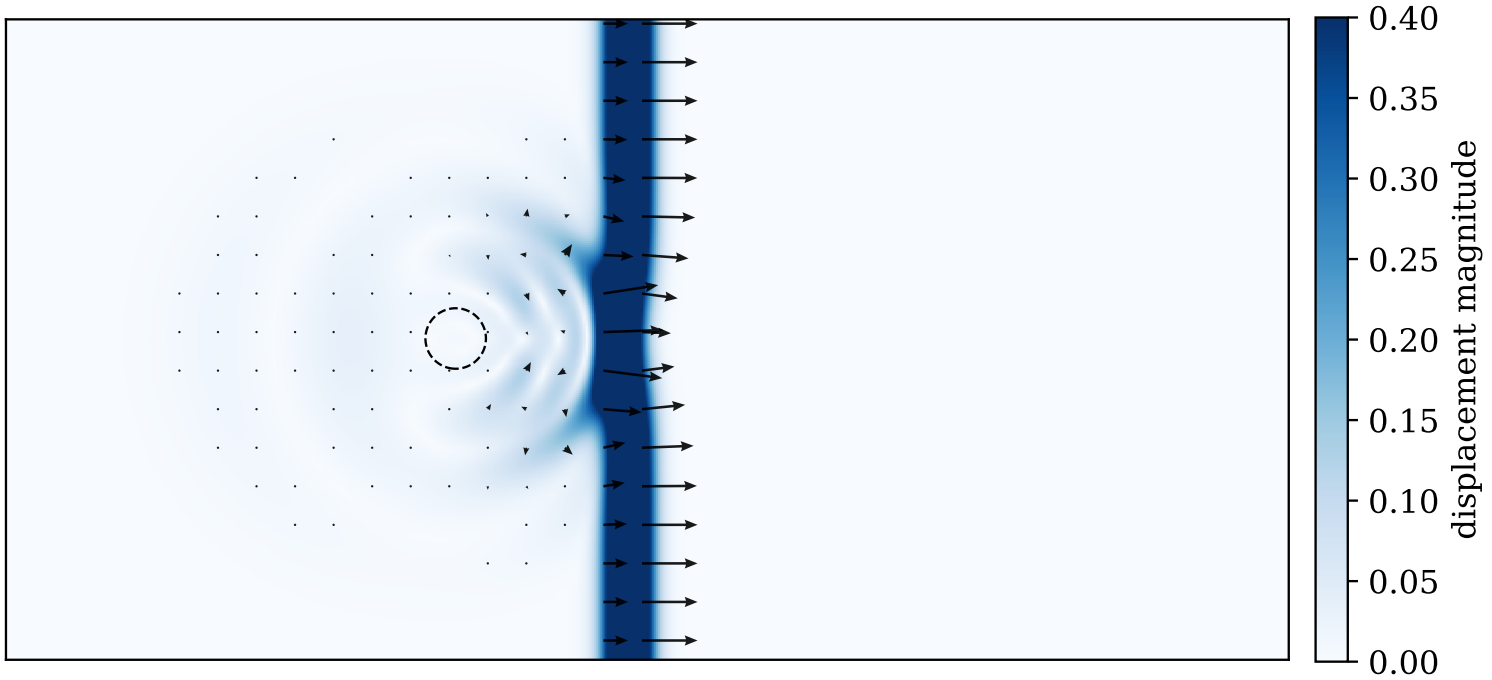
$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{Am} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{Am} - \mathbf{d}) + \frac{\lambda}{2}\mathbf{m}^T\mathbf{Bm}.$$



Wavefront healing as a plane wave passes a small heterogeneity.

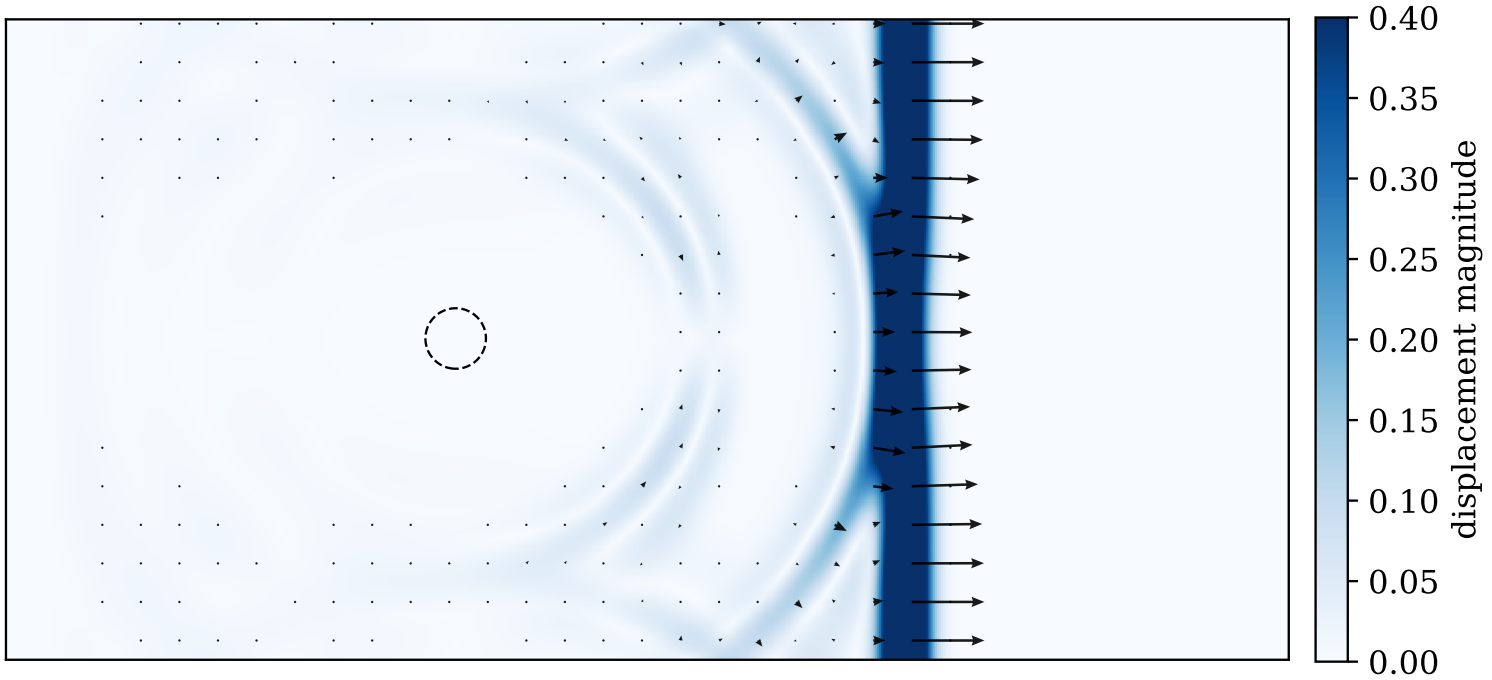
Scattering regime: $\frac{L}{\lambda} \approx 1$

Geometric optics regime: $\frac{L}{\lambda} \gg 1$



Scattering regime: $\frac{L}{\lambda} \approx 1$

Geometric optics regime: $\frac{L}{\lambda} \gg 1$

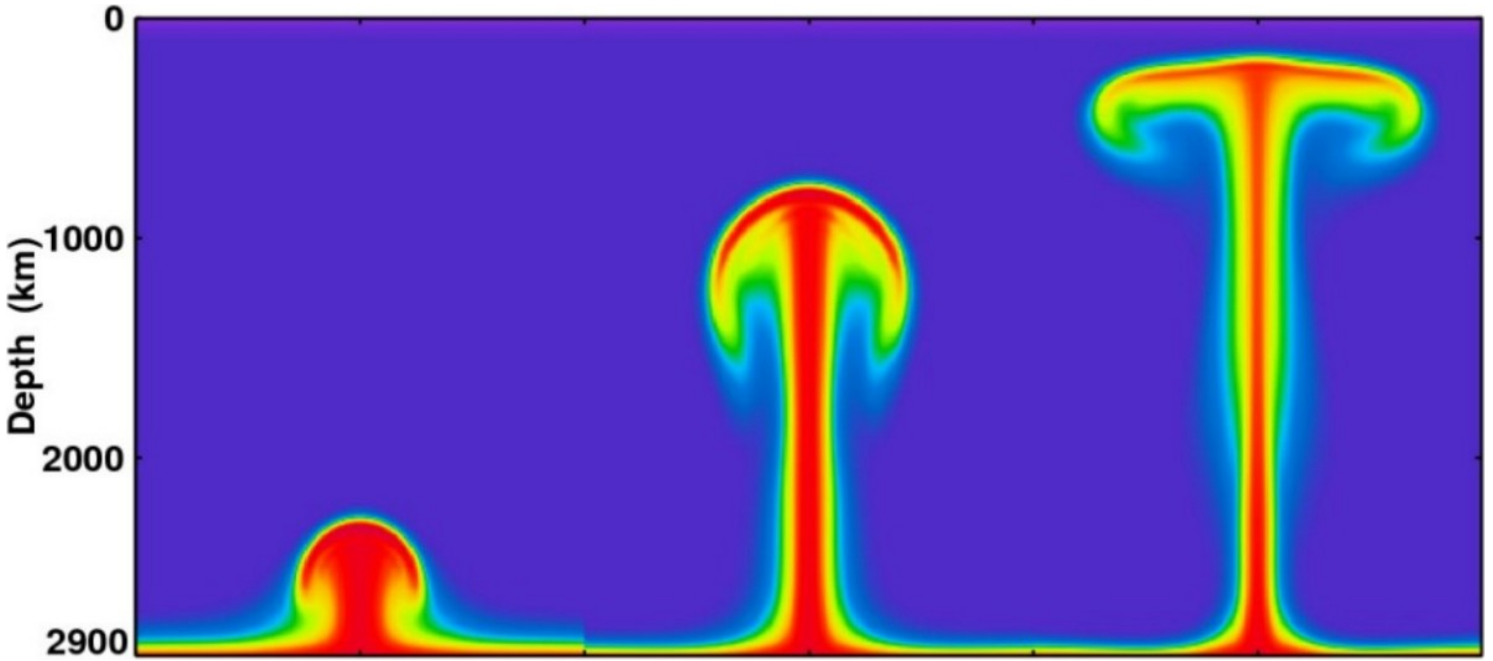


Scattering regime: $\frac{L}{\lambda} \approx 1$

Geometric optics regime: $\frac{L}{\lambda} \gg 1$

Typical periods for global seismology: 1–40 s.

Corresponding wavelengths: 10–400 km.



Farnetani (1997)

Waveform inversion

Equations of motion:

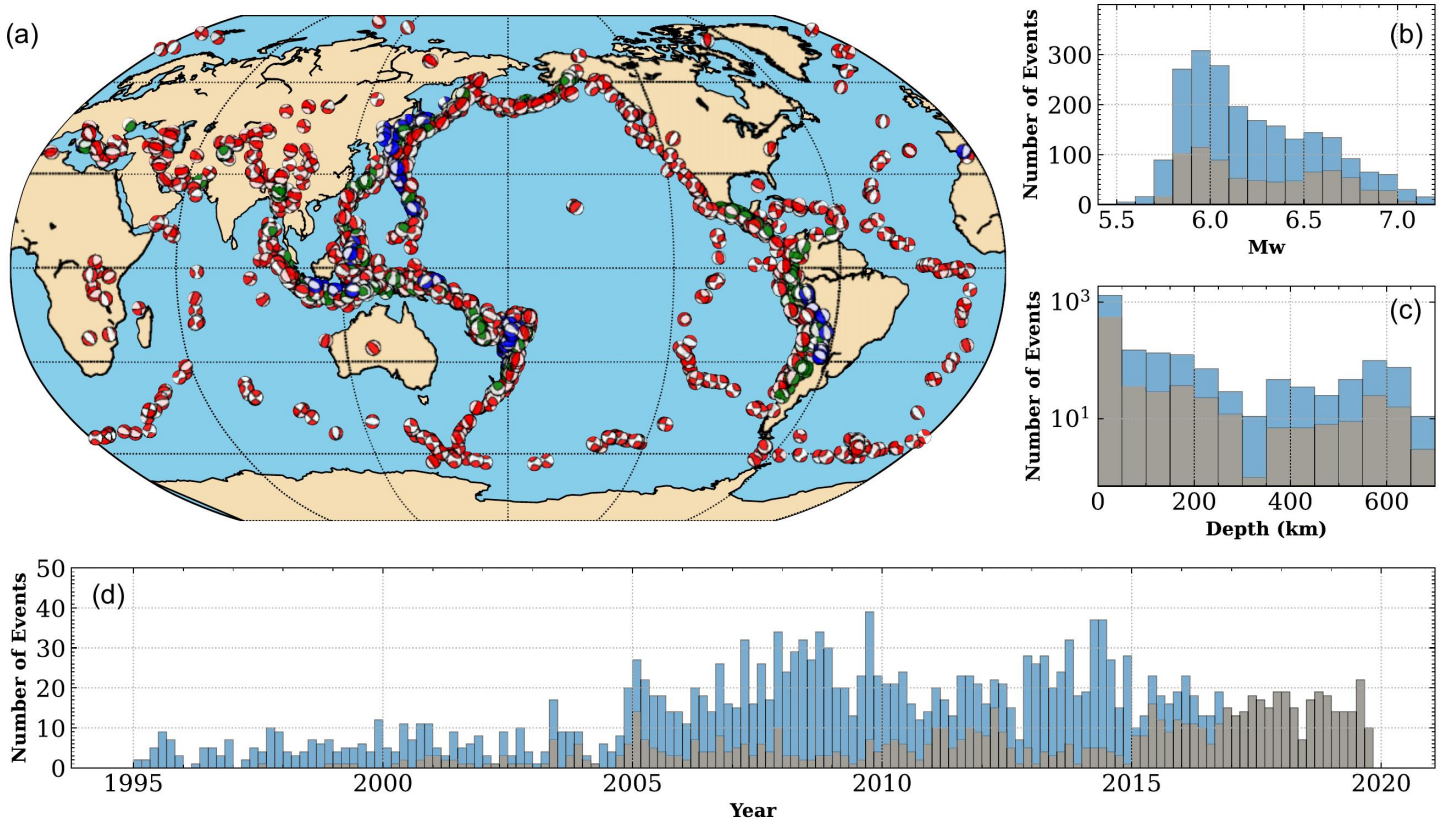
$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) = 0,$$

$$\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \hat{n}_j = 0.$$

These can be solved numerically in realistic laterally heterogeneous earth models using techniques like the **finite element method**.

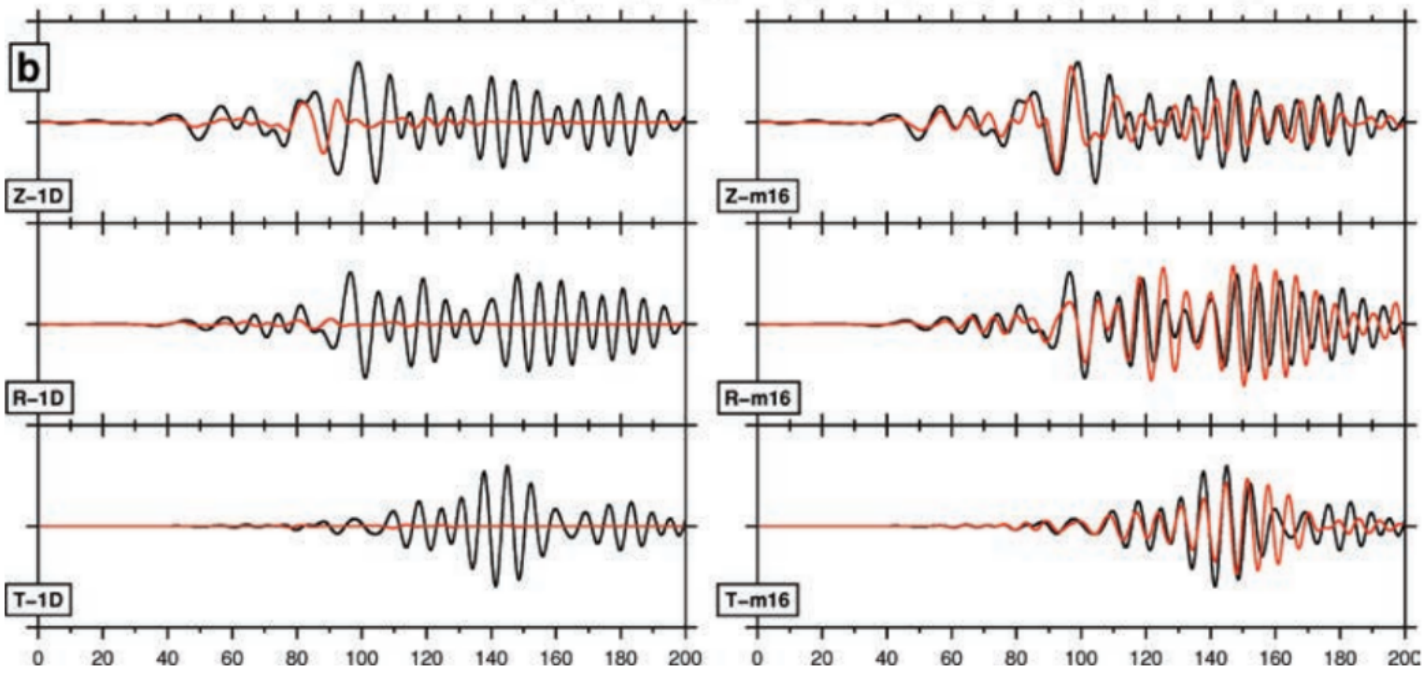
These are demanding calculations, run on HPC systems using parallel computing over hundreds or thousands of cores. GPU computing is now common.

Each simulation takes **tens of minutes to hours** depending on resources.



A typical data set: around 10^3 earthquakes recorded at thousands of stations, each earthquake requiring its own simulation.

Cui *et al.* (2024)



$$J = \frac{1}{2} \int_0^T \|\mathbf{u}(\mathbf{x}_r, t) - \mathbf{u}^{\text{obs}}(t)\|^2 \, \mathrm{d}t.$$

Tape *et al.* (2010)

Misfit is an implicit function of the parameters in the elastic wave equation:

1) **Density** – piecewise smooth function
2) **Elastic tensor** – piecewise smooth tensor field (21 independent components)
3) **Stress glut** – space and time dependent tensor field
4) **Geometry** of the model (e.g. topography on internal boundaries)

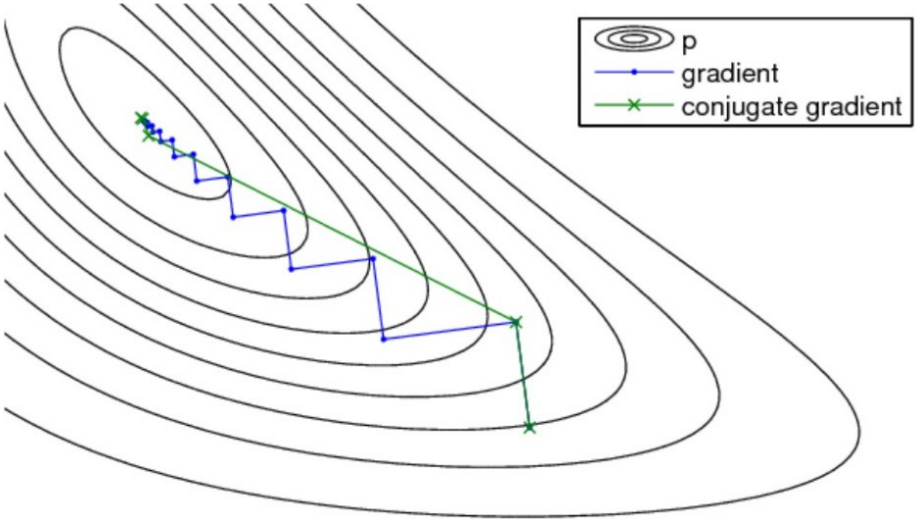
Relation can be condensed to:

$$\mathbf{m} \mapsto J(\mathbf{m}).$$

The aim is to **minimise** this functional.

Strictly defined on an **infinite-dimensional space**. In practice reduced to a finite but high-dimensional space linked to the numerical discretisation of the equations of motion.

Millions or billions of degrees of freedom would be typical.



Iterative reduction of the misfit using the local gradient to take steps downhill.

For large-scale optimisation problems, this is often the only viable approach.

Bayesian sampling methods are not typically an option, though this is an active research topic.

To proceed, we need a way of evaluating the gradients.

Functional derivative defined through a first-order Taylor expansion:

$$J(\mathbf{m} + \delta\mathbf{m}) = J(\mathbf{m}) + \langle DJ(\mathbf{m}) | \delta\mathbf{m} \rangle + O(\|\delta\mathbf{m}\|^2).$$

The terms **gradient** and **sensitivity kernel** are equivalent and widely used.

Focusing on S-wave velocity variations alone, we can write to first order:

$$\langle DJ(\mathbf{m}) | \delta\mathbf{m} \rangle = \int_M K_\beta \, \delta\beta \, \mathrm{d}^3\mathbf{x}.$$

Methods for calculation will be discussed later.

Condition for a (local) minimum, and the condition that follows from it:

$$J(\overline{\mathbf{m}} + \delta\mathbf{m}) \geq J(\overline{\mathbf{m}}),$$

$$DJ(\overline{\mathbf{m}}) = \mathbf{0}.$$

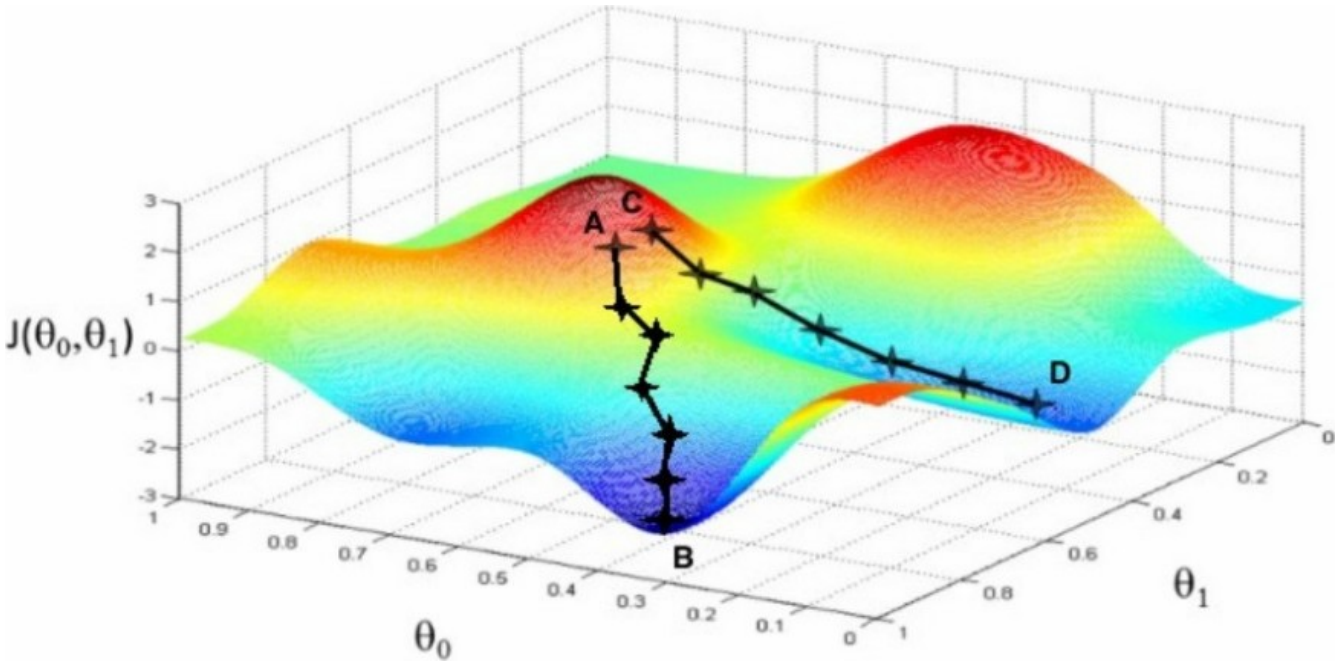
Starting at an initial point, we consider the update:

$$\mathbf{m}_1 = \mathbf{m}_0 - \lambda\, DJ(\mathbf{m}_0).$$

Step length determined by approximately solving a 1D minimisation problem:

$$f(\lambda) = J[\mathbf{m}_0 - \lambda\, DJ(\mathbf{m}_0)].$$

Process is repeated until convergence or the misfit is sufficiently small. This is the **method of steepest descent**.



Gradient-based methods produce only **local minima**.

Uncertainty quantification possible to a limited extent, but adds significantly to the cost.

Calculating the gradients

Finite-difference approximation:

$$\langle DJ(\mathbf{m}) | \delta\mathbf{m} \rangle \approx J(\mathbf{m} + \delta\mathbf{m}) - J(\mathbf{m}).$$

Approximation of the gradient relative to an m-dimensional **model parameterisation** requires $m + 1$ **forward calculations**.

One gradient calculation using this method for a realistic problem would take of order hundreds of years on a massive HPC system!

The **adjoint method** provides the exact gradient at a cost of **two forward simulations**.

We want to minimise:

$$J = \frac{1}{2} \int_0^T \left\| \mathbf{u}(\mathbf{x}_r, t) - \mathbf{u}^{\mathrm{obs}}(t) \right\|^2 \mathrm{d}t,$$

subject to the constraints:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) = 0,$$

$$\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \hat{n}_j = 0.$$

This can be done using the **method of Lagrange multipliers**.

We start by defining the **Lagrangian**:

$$L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = J(\mathbf{u}) - \int_0^T \int_M \left[\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t$$
$$- \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t.$$

Each constraint is associated with a **Lagrange multiplier field**.

Zero initial conditions on displacement and velocity are being imposed directly.

Within the Lagrangian, all fields are regarded as being independent.

The **Lagrange multiplier theorem** states that the following equality holds:

$$DJ(\mathbf{m}) = D_{\mathbf{m}} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}),$$

so long as the following conditions are met:

$$\begin{aligned} D_{\mathbf{u}} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) &= 0, \\ D_{\mathbf{u}'} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) &= 0, \\ D_{\mathbf{w}'} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) &= 0. \end{aligned}$$

Subscripts used for partial functional derivatives.

You only need to know how to apply this method, **not to prove** that it works – details are provided in an appendix to the handout for anyone interested.

Recall that the Lagrangian is:

$$L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = J(\mathbf{u}) - \int_0^T \int_M \left[\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t$$

$$- \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t.$$

Stationarity with respect to $\mathbf{u}'$ gives the equations of motion.

Stationarity with respect to $\mathbf{w}'$ gives the boundary conditions.

This is how the **forward problem** emerges within the method.

Varying with respect to $\mathbf{u}$ gives:

$$\begin{aligned}\langle D_{\mathbf{u}} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) | \delta \mathbf{u} \rangle = \langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle \\ - \int_0^T \int_M \left[\rho \frac{\partial^2 \delta u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t \\ - \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t.\end{aligned}$$

From the chosen form for J we can write:

$$\langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle = \int_0^T \int_{\partial M} h_i' \delta u_i \, \mathrm{d}S \, \mathrm{d}t,$$

$$h_i'(\mathbf{x}, t) = [u_i(\mathbf{x}_r, t) - u_i^{\mathrm{obs}}(t)] \delta(\mathbf{x} - \mathbf{x}_r).$$

We need the following condition to hold:

$$\int_0^T \int_{\partial M} h_i' \delta u_i \, \mathrm{d}S \, \mathrm{d}t - \int_0^T \int_M \left[\rho \frac{\partial^2 \delta u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t$$

$$- \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t = 0,$$

for all variations of the displacement vector.

To see the consequence, we need to pass all the derivatives to the other terms using integration by parts in time and the divergence theorem spatially.

Full details are in the handout.

The result of the long calculation is that:

$$
\begin{aligned}
&- \int_0^T \int_M \left[\rho \frac{\partial^2 u_i'}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k'}{\partial x_l} \right) \right] \delta u_i \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t \\
&- \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j (w_i' - u_i') \, \mathrm{d}S \, \mathrm{d}t \\
&+ \int_0^T \int_{\partial M} \left[h_i' - \left(A_{ijkl} \frac{\partial u_k'}{\partial x_l} \right) \hat{n}_j \right] \delta u_i \, \mathrm{d}S \, \mathrm{d}t \\
&- \int_M \rho \left\{ \left[\frac{\partial \delta u_i}{\partial t} u_i' \right]_{t=T} - \left[\delta u_i \frac{\partial u_i'}{\partial t} \right]_{t=T} \right\} \mathrm{d}^3\mathbf{x} = 0.
\end{aligned}
$$

From this we can read off the necessary adjoint equations.

We have shown that the Lagrange multipliers must satisfy the equations:

$$\rho \frac{\partial^2 u_i'}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k'}{\partial x_l} \right) = 0,$$

$$\left(A_{ijkl} \frac{\partial u_k'}{\partial x_l} \right) \hat{n}_j = h_i', \qquad u_i'(\mathbf{x}, T) = \frac{\partial u_i'}{\partial t}(\mathbf{x}, T) = 0,$$

along with the equality $\mathbf{w}' = \mathbf{u}'$ on the boundary.

This is a **terminal-value problem** for $\mathbf{u}'$ driven by the boundary traction h_i':

$$h_i'(\mathbf{x}, t) = [u_i(\mathbf{x}_r, t) - u_i^{\mathrm{obs}}(t)]\delta(\mathbf{x} - \mathbf{x}_r).$$

The terminal-value problem must be solved backwards in time from $t = T$. Instead, define the time-reversed field:

$$u_i^\dagger(\mathbf{x}, t) = u_i'(\mathbf{x}, T - t).$$

This satisfies the same wave equation, with boundary traction and initial conditions:

$$\left(A_{ijkl} \frac{\partial u_k^\dagger}{\partial x_l} \right) \hat{n}_j = h_i^\dagger, \quad h_i^\dagger(\mathbf{x}, t) = h_i'(\mathbf{x}, T - t),$$

$$u_i^\dagger(\mathbf{x}, 0) = \frac{\partial u_i^\dagger}{\partial t}(\mathbf{x}, 0) = 0.$$

An **initial-value problem** of the same form as the forward problem, solved with the same numerical code.

Time reversal is just a device for converting a terminal-value problem into an initial-value one.

Using the multiplier theorem and integrating by parts, we find that:

$$\langle DJ(\mathbf{m})|\delta\mathbf{m}\rangle = \int_0^T \int_M \left[\delta\rho \frac{\partial u_i}{\partial t}\frac{\partial u_i'}{\partial t} - \delta A_{ijkl}\frac{\partial u_i'}{\partial x_j}\frac{\partial u_k}{\partial x_l} + \delta\overline{S}_{ij}\frac{\partial u_i'}{\partial x_j}\right] \mathrm{d}^3\mathbf{x}\,\mathrm{d}t.$$

We can then read off the **sensitivity kernels**:

$$K_\rho(\mathbf{x}) = \int_0^T \frac{\partial u_i}{\partial t}(\mathbf{x},t)\frac{\partial u_i'}{\partial t}(\mathbf{x},t)\,\mathrm{d}t,$$

$$K_{A_{ijkl}}(\mathbf{x}) = -\int_0^T \frac{\partial u_i'}{\partial x_j}(\mathbf{x},t)\frac{\partial u_k}{\partial x_l}(\mathbf{x},t)\,\mathrm{d}t,$$

$$K_{\overline{S}_{ij}}(\mathbf{x},t) = \frac{\partial u_i'}{\partial x_j}(\mathbf{x},t).$$

In terms of the time-reversed field, the kernels read:

$$K_\rho(\mathbf{x}) = -\int_0^T \frac{\partial u_i}{\partial t}(\mathbf{x}, t)\frac{\partial u_i^\dagger}{\partial t}(\mathbf{x}, T-t)\,\mathrm{d}t,$$

$$K_{A_{ijkl}}(\mathbf{x}) = -\int_0^T \frac{\partial u_i^\dagger}{\partial x_j}(\mathbf{x}, T-t)\frac{\partial u_k}{\partial x_l}(\mathbf{x}, t)\,\mathrm{d}t,$$

$$K_{\overline{S}_{ij}}(\mathbf{x}, t) = \frac{\partial u_i^\dagger}{\partial x_j}(\mathbf{x}, T-t).$$

The forward field at time t is paired with the adjoint field after it has evolved for a time $T-t$.

Here $\mathbf{u}^\dagger$ is the **adjoint wavefield** and $h_i^\dagger$ the **adjoint traction** (or **adjoint source**).

For each gradient calculation we must:

1) Solve the forward problem for $\mathbf{u}$
2) From this solution, compute J and form the adjoint traction $h_i^\dagger$
3) Solve the adjoint problem for $\mathbf{u}^\dagger$ as an initial-value problem
4) Combine the forward and adjoint fields to form the sensitivity kernels

Such a calculation has the cost of two forward calculations (three in practice, so that the forward field need not be stored).

The adjoint equations are identical in form to the forward ones, and so the same numerical code can be used.

Choice of misfit enters into the problem only through the form of the adjoint traction.

Calculations that would take hundreds of years are reduced to minutes or hours.

In practice, delay times are measured using **cross-correlations** of windowed waveforms:

$$s^{\mathrm{obs}}(t) = w(t)\hat{v}_i u_i^{\mathrm{obs}}(t), \qquad s(\mathbf{x}_r, t) = w(t)\hat{v}_i u_i(\mathbf{x}_r, t),$$

$$C(\tau) = \int_{-\infty}^{\infty} s^{\mathrm{obs}}(t-\tau) s(\mathbf{x}_r, t)\, \mathrm{d}t.$$

The time shift $\overline{\tau}$ that maximises the cross-correlation is taken as the delay time.

A late observed arrival gives $\overline{\tau} < 0$, the opposite sign to the delay time of Lecture 19. Only the perturbation $\delta\overline{\tau}$ enters the kernels, so nothing depends on this choice.

This defines a functional of the synthetic waveform, and hence we can derive the associated sensitivity kernels using the adjoint method.

Perturbing the condition $\mathrm{d}C/\mathrm{d}\tau(\overline{\tau}) = 0$, with the waveforms differing only by the time shift:

$$\delta\overline{\tau} = -\frac{1}{N}\int_{-\infty}^{\infty} \dot{s}(\mathbf{x}_r, t)\, \delta s(\mathbf{x}_r, t)\, \mathrm{d}t, \quad N = \int_{-\infty}^{\infty} \dot{s}(\mathbf{x}_r, t)^2\, \mathrm{d}t.$$

From this we read off the adjoint traction:

$$h_i'(\mathbf{x}, t) = -\frac{1}{N} w(t) \dot{s}(\mathbf{x}_r, t) \hat{v}_i \delta(\mathbf{x} - \mathbf{x}_r).$$

The observed seismogram has dropped out: the kernel depends only on the reference model.

For the cross-correlation delay time, we obtain to first order:

$$\delta\overline{\tau} = \int_M \left(K_\alpha\,\delta\alpha + K_\beta\,\delta\beta\right)\mathrm{d}^3\mathbf{x}.$$

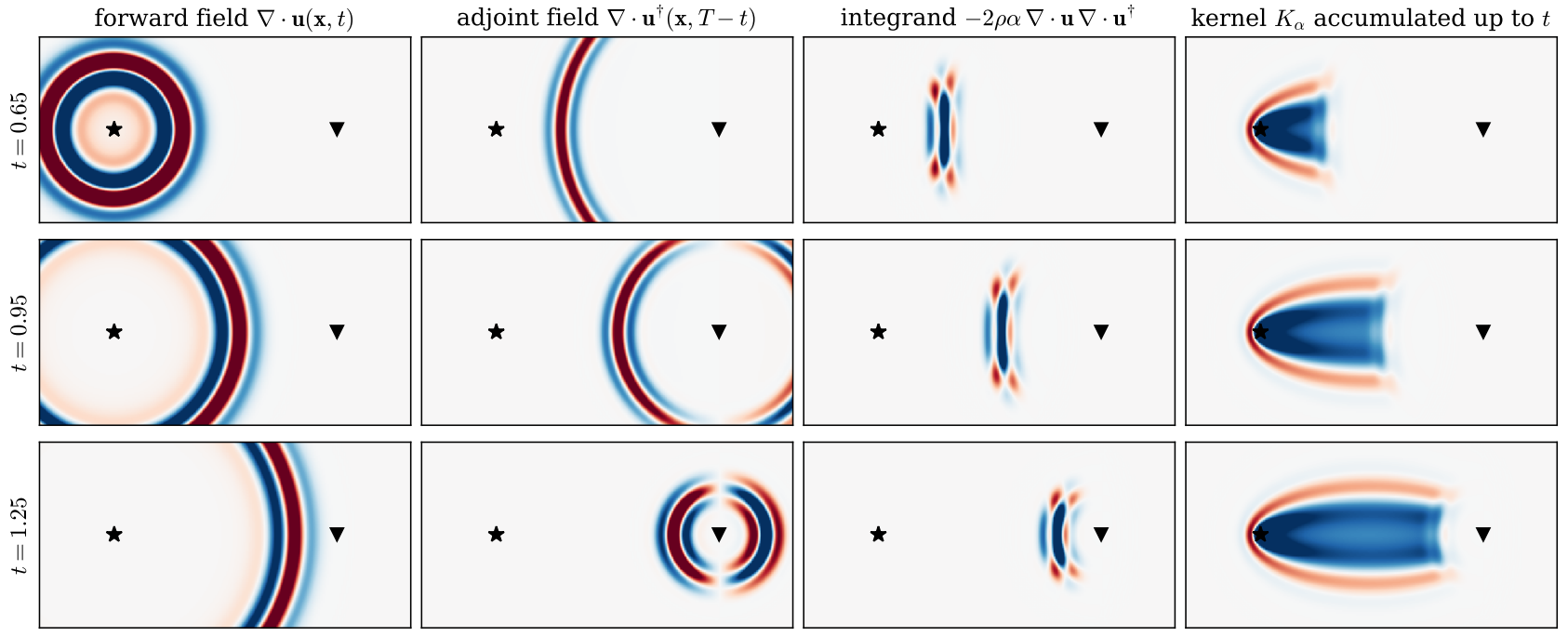
This can be compared with the ray-theoretic result from Fermat's principle:

$$\delta T = -\int_{\mathrm{ray}} \frac{\delta c}{c^2}\,\mathrm{d}s.$$

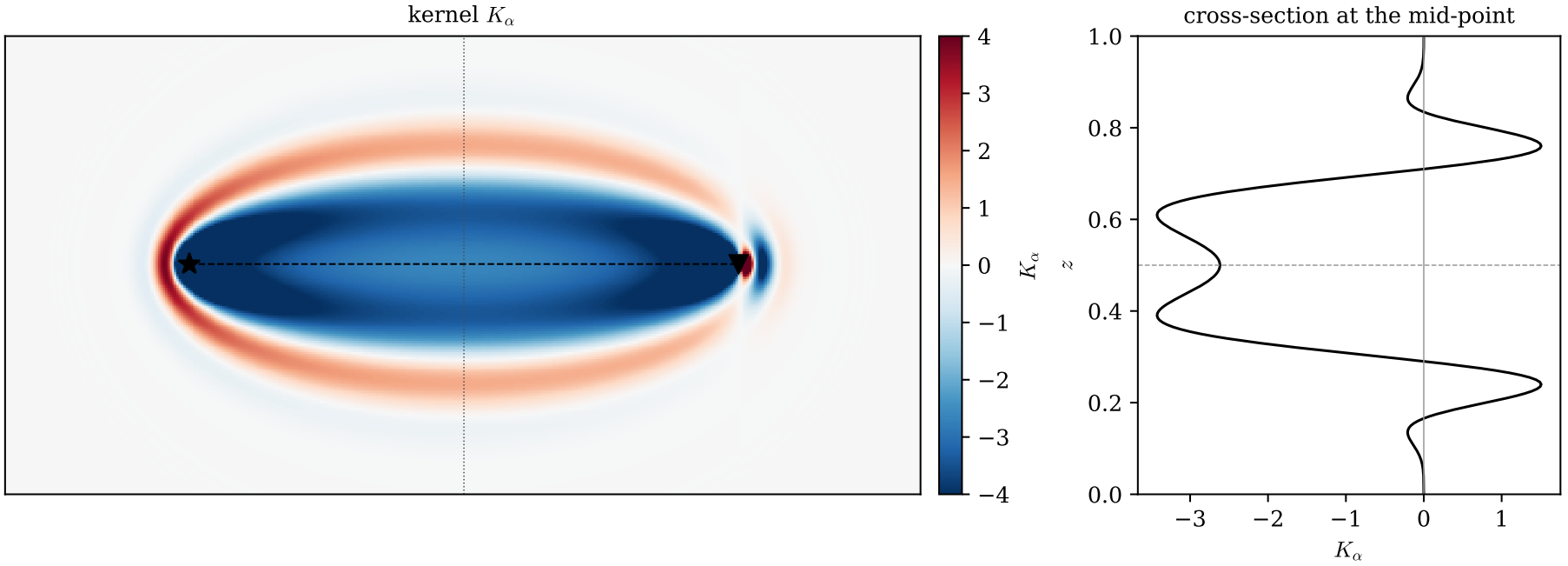
Within ray theory all the sensitivity lies on the ray path. What does the adjoint method give?

Building a P-wave kernel by the adjoint method in two dimensions:

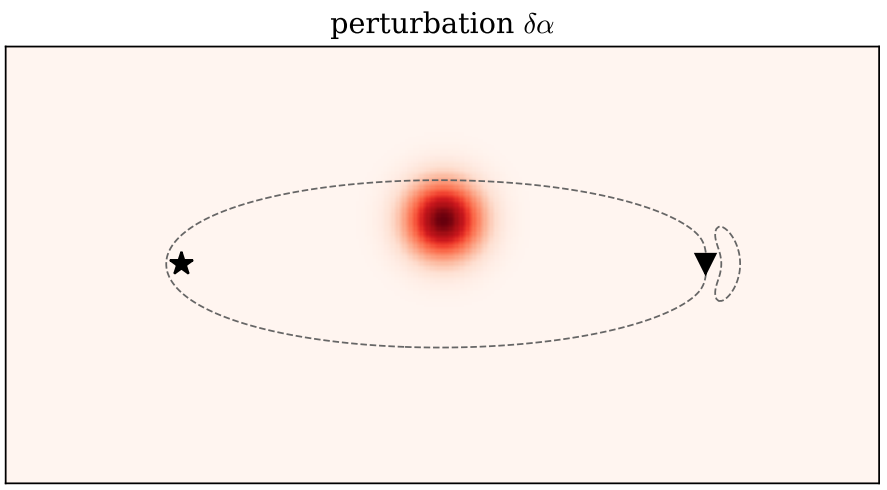
$$K_\alpha(\mathbf{x}) = -2\rho\alpha \int_0^T \nabla \cdot \mathbf{u}(\mathbf{x}, t)\, \nabla \cdot \mathbf{u}^\dagger(\mathbf{x}, T-t)\, \mathrm{d}t.$$

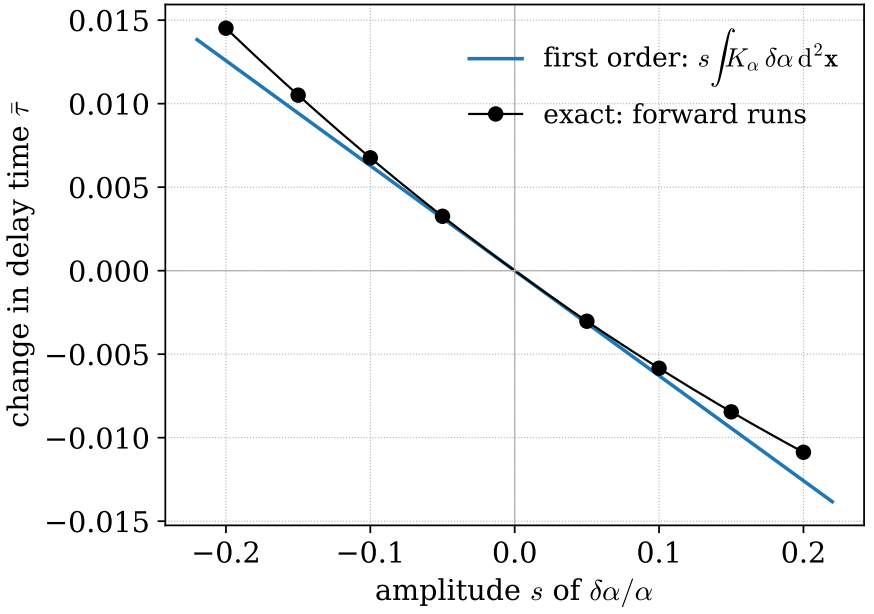


Sensitivity accumulates only where the forward field at time t overlaps the adjoint field at time $T - t$.

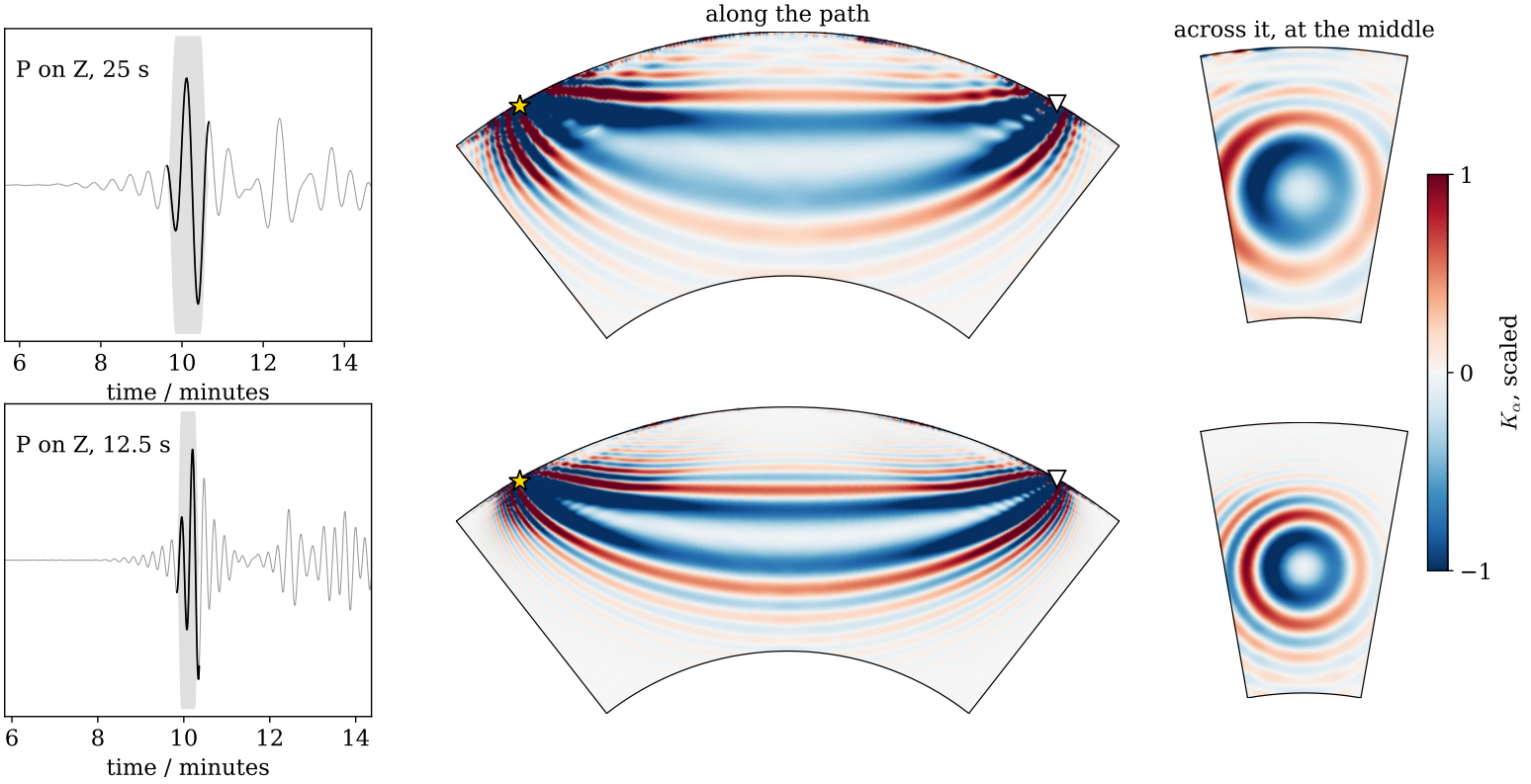


The kernel is negative throughout the **first Fresnel zone**, since an increase in wave speed there advances the arrival, and changes sign in the surrounding zones.



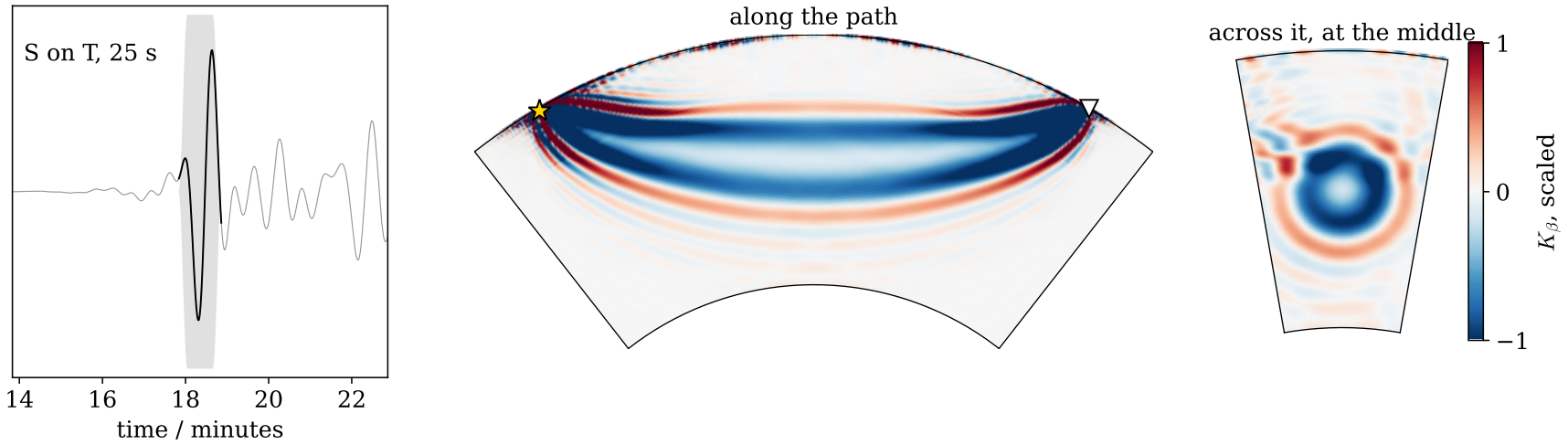


A **Taylor test**: a fixed perturbation $\delta\alpha$ is scaled by s, and the change in the delay time found by rerunning the forward problem is compared with the prediction from the kernel. They agree to first order in s, the difference shrinking like s^2.



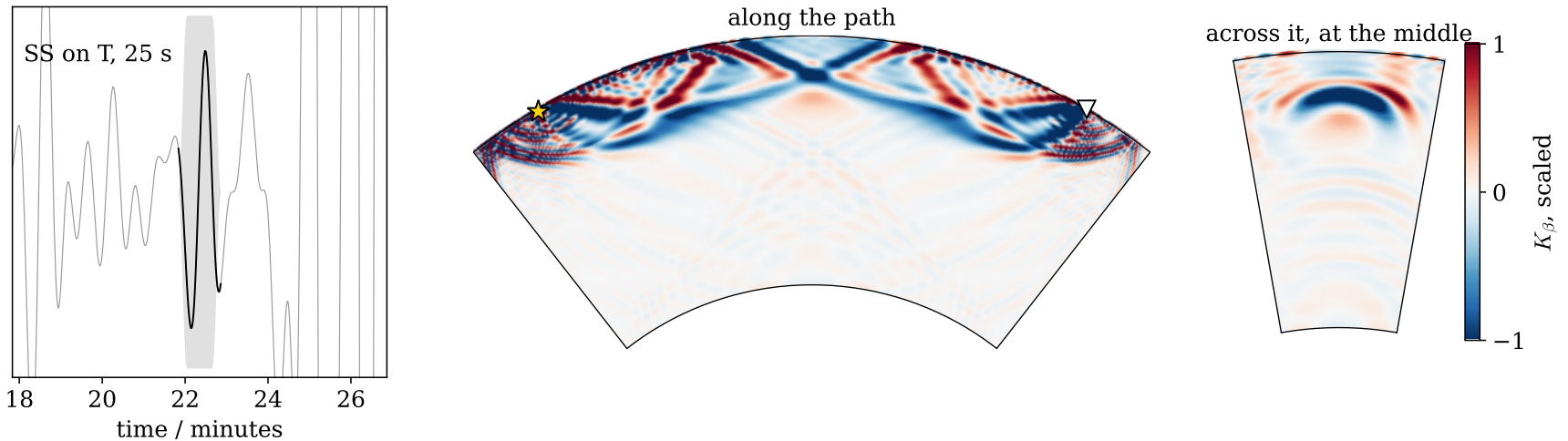
In three dimensions the sensitivity lies within the first Fresnel zone, which narrows at shorter periods, but curiously vanishes on the ray path itself. P-wave traveltime at 60° in PREM, measured in the window shaded.

Calculated with YSpec



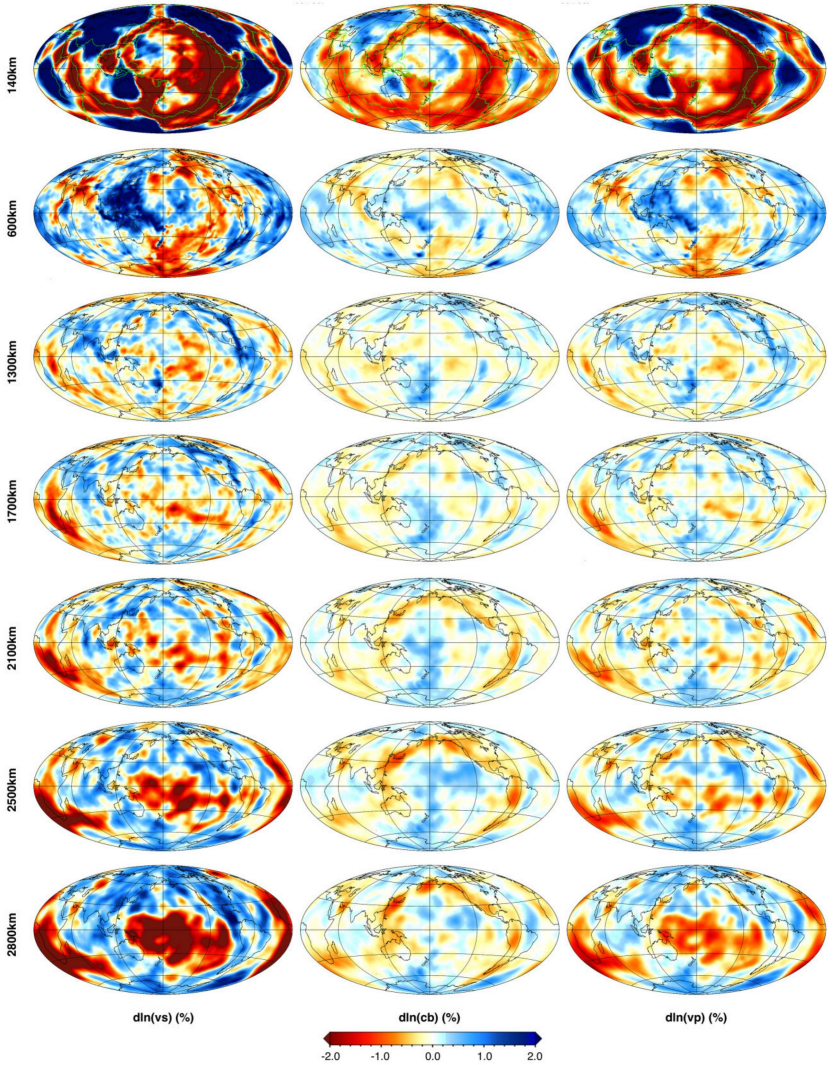
Named **banana doughnut kernels** by Tony Dahlen (1942–2007). S-wave traveltime at 60° in PREM.

Calculated with YSpec



The kernel of SS, reflected at the surface midway between the source and the receiver, is far from a single banana.

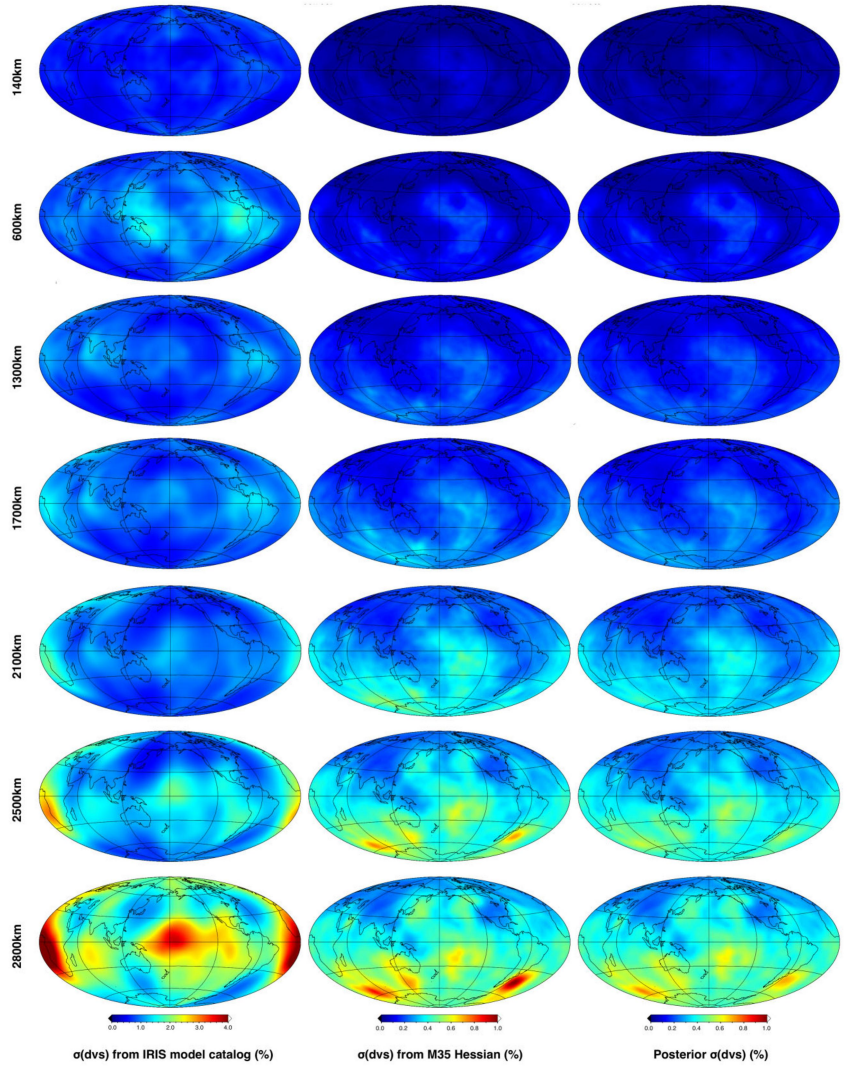
Calculated with YSpec



Cui *et al.* (2024)

GLAD-M35: shear wave speed, bulk sound speed and compressional wave speed at depths from 140 to 2800 km.

Waveforms from 2160 earthquakes at 11 575 stations, and 35 iterations of gradient-based optimisation.

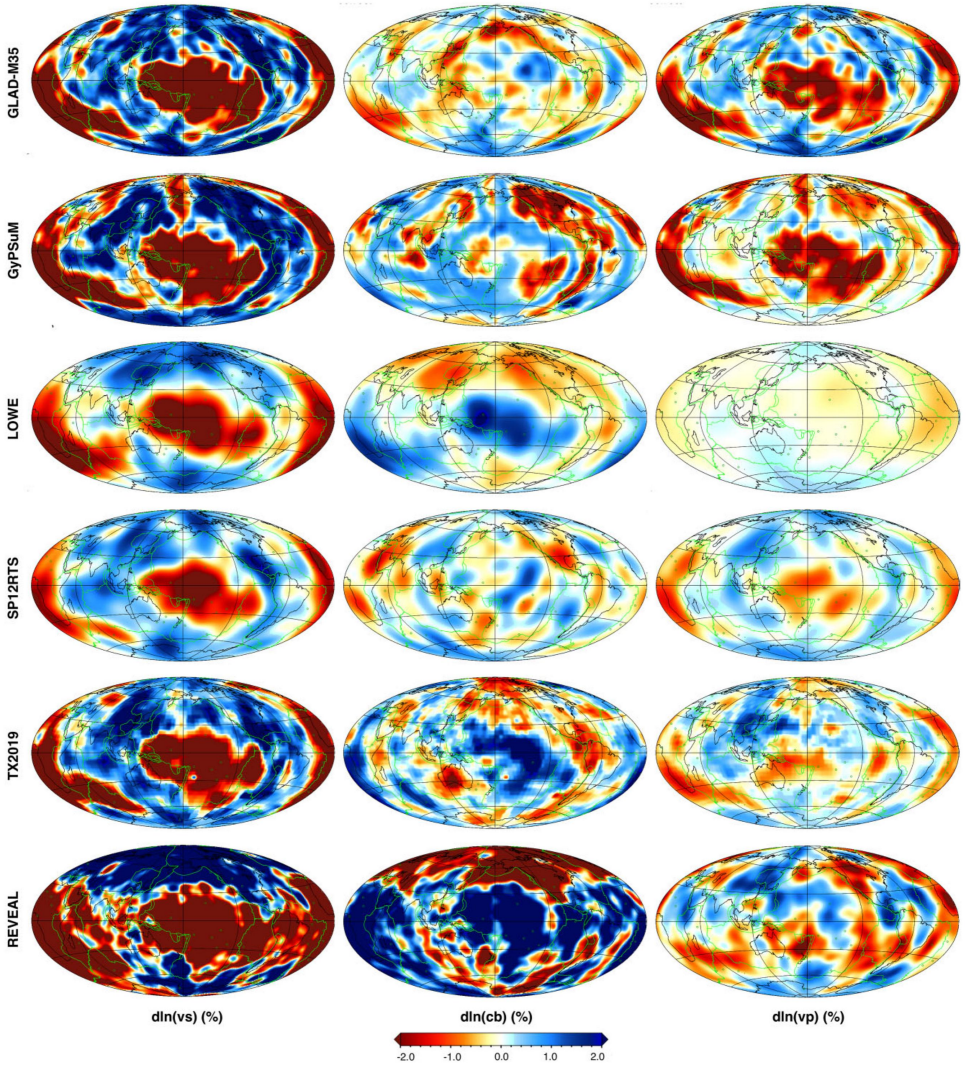


Cui *et al.* (2024)

Uncertainty can be estimated approximately: the Hessian of the misfit plays the role of the inverse posterior covariance.

Its action on a vector costs a few further forward and adjoint simulations.

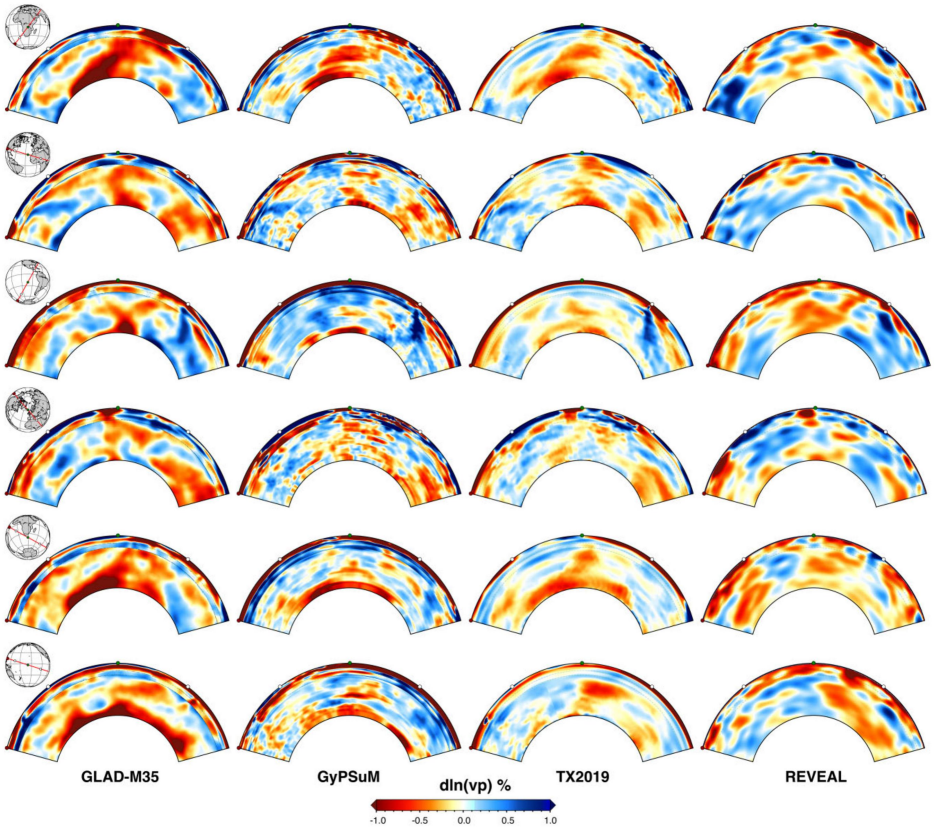
Uncertainty is reduced most where the data coverage is best.



Six global models at 2800 km depth, three of them from waveform tomography.

They agree on the largest features, but differ considerably in detail, notably in the bulk sound speed.

Cui *et al.* (2024)



Cross-sections beneath hot spots: the models differ markedly in whether they show continuous low-velocity conduits, or plumes, rising through the mantle.

Cui *et al.* (2024)

What you need to know and be able to do

(i) That waveform tomography depends on the ability to efficiently solve the elastic wave equation in realistic earth models.

(ii) How least-squares waveform misfits can be defined, and what is meant by their gradients.

(iii) How gradient-based optimisation works, why it is useful, and what its limitations are.

(iv) How the adjoint method can be used to calculate gradients, and what its advantages are. The derivation in this lecture is too lengthy for you to reproduce in full. But you may be asked to perform parts of the calculations with suitable guidance or to apply the method to simpler systems.